Synthetic Dental X-Ray Generation and Segmentation Analysis

Computer Vision Exam

Alessia Arena
Alessandro Marciano

Main objectives

Train and finetune a GAN to generate realistic orthopanoramic x-ray images → **Synthetic Data Generation**

Train a CNN-based segmentation model and **finetune** the model to correctly identify and segment teeth structures. → **Teeth Segmentation**

Test the trained segmentation model on the synthetic dataset and **compare its performance** with real X-ray images. → **Evaluation and Generalization Study**

Evaluate different GAN architectures and assess segmentation performance in various augmentation scenarios. → **Ablation Study**

Proposed approach

Kaggle was selected as the primary training environment for this project due to its highly advantageous free-tier resources. Specifically, the platform provided access to **dual NVIDIA T4 GPUs with 16 GB of VRAM**, allowing for up to **30 hours of compute time per week**, structured in continuous 12-hour sessions.

Synthetic Data Generation →

- **StyleGAN-ADA2**: Unconditional generation (Synthetic images from scratch)
- **Pix2Pix**: Conditional generation (Image-to-image translation task from segmentation masks)

Teeth Segmentation →

- **YOLOv8-seg**

State of the art - Generation

DIFFUSION MODELS

- **Strengths:**
  - **Absolute SOTA:** Latent Diffusion Models (LDMs) provide unmatched anatomical fidelity and sample diversity.
  - **Solves GAN Issues:** Completely overcomes mode collapse.
  - **Conditional Synthesis:** Diffusion + **ControlNet** surpasses traditional translation models (like Pix2Pix) in edge and texture precision.

LIMITATIONS →

- **High Resource Demand:** Prohibitive computational costs and massive VRAM requirements.
- **Speed Bottleneck:** Slow, iterative inference times compared to single-pass models.

Notable Literature: *PanoDiff-SR*, *ToothMaker*

State of the art - Generation

GENERATIVE ADVERSARIAL NETWORKS (GANs)

- **Strengths:**
  - **Inference Efficiency:** Single-pass generation ensures near real-time synthesis and vastly lower computational overhead compared to iterative diffusion models.
  - **Alias-Free Synthesis:** The current unconditional SOTA, **StyleGAN3**, provides strict translation and rotation equivariance, eliminating the "texture sticking" artifacts that plague older medical GANs.
  - **Advanced Translation:** Frameworks like **SPADE** (Spatially-Adaptive Normalization) and **CUT** (Contrastive Unpaired Translation) represent the peak for high-resolution mask-to-image and cross-modality (e.g., MRI-to-CT) synthesis.

LIMITATIONS →

- **Mode Collapse & Mode Dropping**
- **Training Instability**
- **Anatomical Hallucinations**

State of the art - Segmentation

- **High-Efficiency Single-Stage Architectures (Project Baseline):**
  - **YOLOv8-seg:** Selected as the optimal baseline for hardware-constrained instance segmentation.
  - **YOLO-DentSeg (2025):** Recent literature validates YOLOv8-based architectures for real-time analysis of dental panoramic radiographs.

- **Compute-Intensive Clinical Architectures:**
  - **Semantic Segmentation: nnU-Net** remains the benchmark framework, auto-configuring optimal U-Net topologies for specific medical datasets.
  - **Instance Segmentation: Mask2Former** utilizes Transformer-based attention to delineate overlapping anatomical structures (e.g., individual teeth) with superior edge accuracy.
  - **Foundation Models: MedSAM** provides unprecedented zero-shot generalization capabilities, pre-trained on multi-modal medical datasets.

DATASETS USED

DENTEX:

- **693** X-rays labeled for **quadrant detection and quadrant classes** only.
- **634** X-rays labeled for tooth detection with **quadrant and tooth enumeration classes**.
- **705** X-rays fully labeled for abnormal tooth detection with **quadrant, tooth enumeration, and diagnosis classes**.
- An additional **1571 unlabeled** X-rays.
- **50 validation** images.

UFBA-425:

- A **425**-image subset of the original UFBA dataset, featuring high-precision **polygonal masks** from expert manual re-annotation.

TUFTS:

- A comprehensive collection of **1,000** panoramic radiographs featuring expert-level **annotations for both individual teeth and dental abnormalities**, further enriched by unique radiologist eye-tracking data.

Synthetic Data Generation: GAN choice

StyleGAN2-ADA:

StyleGAN2-ADA was selected for its ability to generate high-fidelity synthetic images from limited datasets. By leveraging **Adaptive Discriminator Augmentation**, the model prevents overfitting on small datasets, ensuring realistic and diverse orthopanoramic X-ray generation despite the relatively small sample size.

Images used for **training**:

- 1000 images from TUFTS,
- 693 from DENTEX Quadrant Detection,
- 454 from DENTEX Quadrant Detection Enumeration,
- 505 from DENTEX Quadrant Detection Enumeration Disease,
- 1571 from DENTEX unlabeled,
- 50 from DENTEX validation

For a total of **4,273 images**

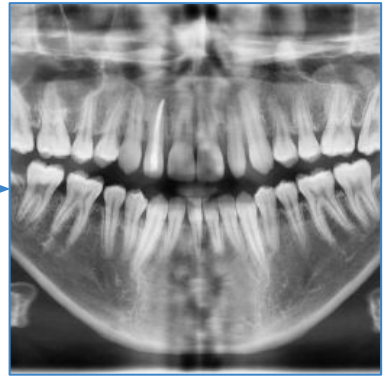
Synthetic Data Generation: GAN choice

DATA STANDARDIZATION & PRE-PROCESSING

- **Squared Cropping:** Targeted removal of non-informative peripheral regions to enforce a strict 1:1 aspect ratio, a structural prerequisite for GAN processing.
- **Resizing:** Downscaling to 256x256 resolution to strictly preserve original anatomical proportions.
- **CLAHE (Contrast Limited Adaptive Histogram Equalization) Application:** Local contrast enhancement to mitigate exposure discrepancies between datasets.
- **Grayscale-to-RGB Conversion:** Channel duplication required for StyleGAN2-ADA transfer

DENTEX image Example

pre-processed



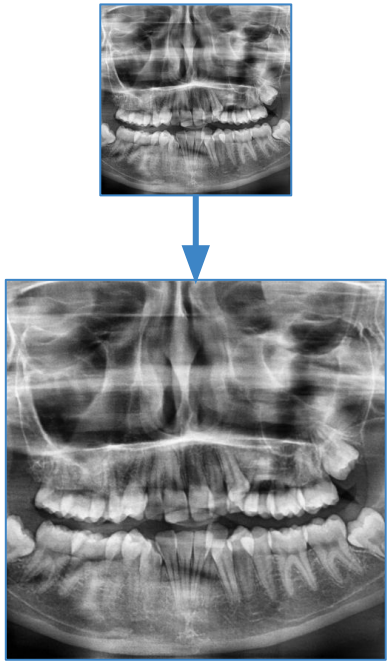
post-processed

Synthetic Data Generation: GAN choice

HARDWARE CONSTRAINTS & SUPER-RESOLUTION PIPELINE

- **High resolution constraints:** Training the GAN directly at 512px caused immediate **CUDA Out of Memory (OOM)** errors, whereas reducing the batch size to fit the memory triggered severe **Mode Collapse**.
- **The Decoupled Solution:** To break this trade-off, image synthesis and high-resolution scaling were separated into a two-stage pipeline.
  - **Stage 1 (Low res generation):** The GAN was trained at **256px**, allowing for optimal batch sizes, ensuring mathematical stability and anatomical diversity.
  - **Stage 2 (Upscaling): Real-ESRGAN** was deployed to upscale the generated images to **512px**, meeting the quality threshold for YOLOv8 segmentation.
- **Custom SR Training:** The Real-ESRGAN model was trained on paired 256px/512px samples from the styleGAN2-ADA dataset to prevent non-anatomical upscaling artifacts.

Real image 256x256



Upscaled 512x512

Synthetic Data Generation: GAN choice

StyleGAN2-ADA results:

Synthetic Data Generation: GAN choice

Pix2Pix:

Pix2Pix was implemented as a conditional GAN to serve as a core **data augmentation strategy**, enhancing the YOLOv8 training process. By treating the generation as an image-to-image translation task, the model synthesizes realistic radiographs **directly from segmentation masks**. This pipeline enables the procedural generation of new masks from existing ones, which are then fed into the Pix2Pix model to obtain novel image-mask pairs, significantly expanding the dataset for segmentation training.

Images used for **training**:

- 454 images from DENTEX Quadrant Detection Enumeration
- 425 from UFBA-425

For a total of **879 images**

Additionally, a separate set of **180 images** from DENTEX was used for **validation**.

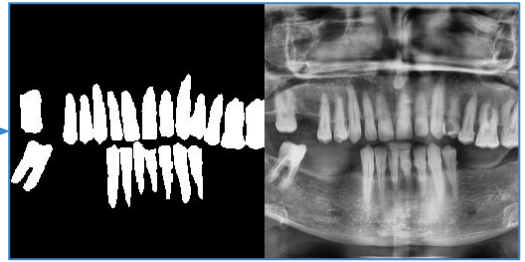
Synthetic Data Generation: GAN choice

DATA STANDARDIZATION AND PRE-PROCESSING

- **Foundation Model Refinement (SAM):** Deployment of the **Segment Anything Model** to precisely delineate anatomical dental contours, standardizing mask geometry across varied datasets and elevating ground-truth quality.
- **Paired Cropping & Resizing:** Synchronous square cropping and downscaling of the images and associated mask to a uniform **256x256** resolution
- **Format Conversion & Rasterization:** Translating raw bounding-box/coordinate annotations (.txt) into standardized, rasterized segmentation masks (.png).
- **Spatial Concatenation:** Structurally aligning and merging each mask side-by-side (left-aligned) with its corresponding radiograph to construct the exact paired input required by Pix2Pix.

DENTEX image Example

pre-processed



post-processed

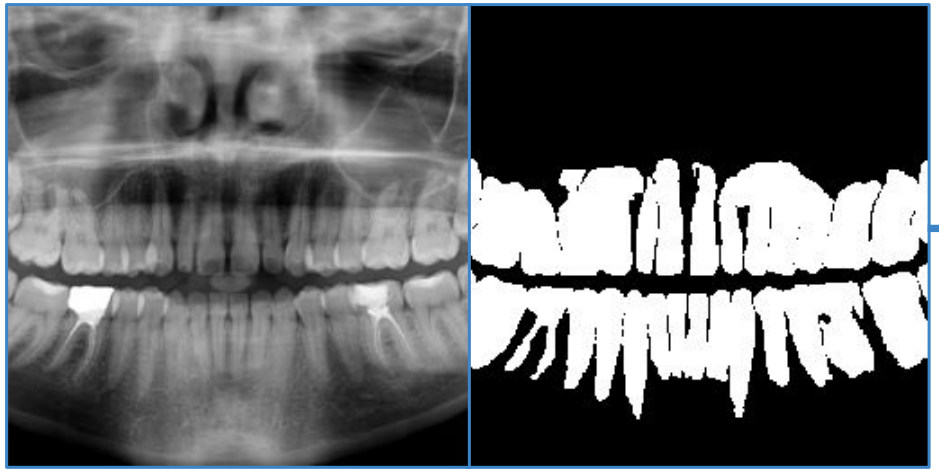
Synthetic Data Generation: GAN choice

HARDWARE CONSTRAINTS AND POSSIBLE BIASES

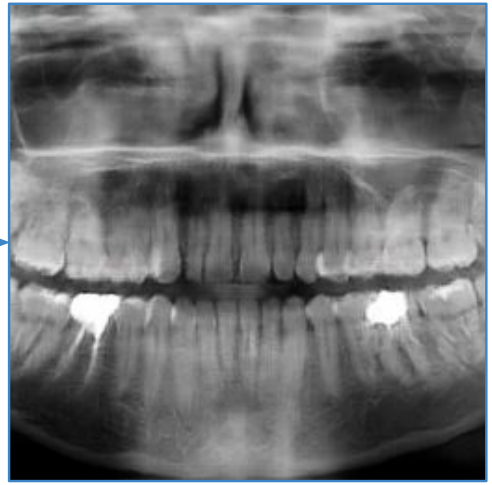
- **Persistent Hardware Constraints:** Consistent with the unconditional GAN, training the Pix2Pix architecture directly at high resolutions exceeded the Kaggle 16GB VRAM limit.
- **Super-Resolution Re-deployment:** To bypass this limitation, Pix2Pix generation was constrained to **256px**. The previously fine-tuned **Real-ESRGAN** model was then seamlessly re-deployed to upscale these conditional outputs to the required **512px** baseline.
- **Dataset Variance & Domain Bias:** Variations in inherent contrast and anatomical proportions between the DENTEX and UFBA datasets caused the model to partially overfit to these distinct source distributions.
- **Conditional Style Entanglement:** Consequently, the generated image style became correlated with the input mask's geometry, producing outputs that visually mirrored either the DENTEX or the UFBA domain based on the mask's proportions.
- **Preservation of Utility:** Despite this stylistic bias, the structural integrity and anatomical realism of the generated radiographs remained fully intact, successfully validating their use as possible data augmentation samples.

Synthetic Data Generation: GAN choice

PIX2PIX RESULTS



Real Image and its associated mask



pix2pix generated image

Teeth Segmentation

Segmentation with YOLOv8-seg: Architecture and Validation

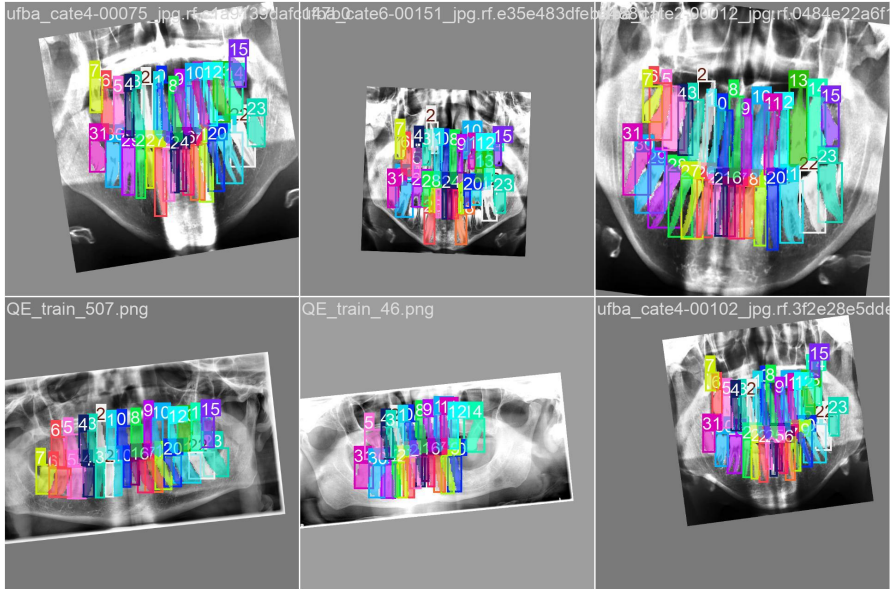
- **Literature-Validated Architecture:** YOLOv8-seg offers optimal balance between accuracy and speed when analyzing complex dental X-rays, as seen in *YOLO-DentSeg* framework.
- **Anchor-Free Approach for Medical Geometries:** By dynamically predicting the object's center, it overcomes rigid BBox definitions and ensures adaptation to diverse dental shapes and inclinations.
- **Unified Approach (Detection + Masking):** Simultaneously executes tooth localization and pixel-level mask generation, while maintaining clinical-grade precision, allowing for mask overlapping.
- **Maximum Synergy with GAN Augmentation:** The network introduces its own data augmentation, effectively translating into high generalization capabilities, even when synthetic images are introduced in the dataset.

The same dataset used for the Pix2Pix GAN was employed due to the limited availability of labeled X-rays, totaling **879 images**. Nevertheless, it was sufficient for Quadrant and Enumeration teeth detection.

Additional tests with Pix2Pix synthetic images were carried out, expanding the dataset up to **1758 images**.

Teeth Segmentation

YOLOv8-seg results:



TRAIN BATCH

VALIDATION BATCH

Evaluation and Generalization study

YOLOv8 trained model was tested over **4 different 380-images sets**:

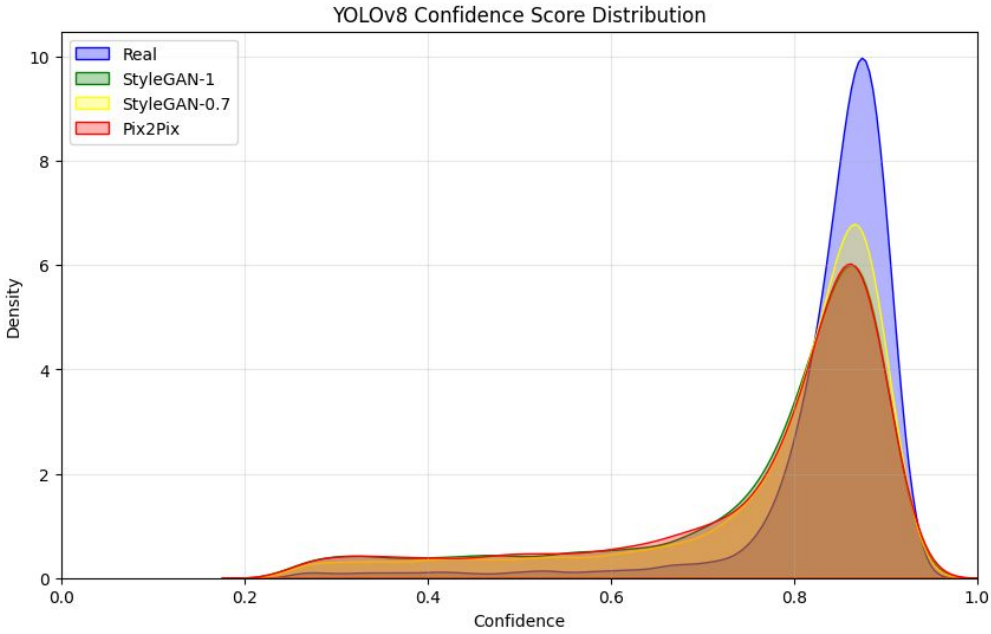
- Real
- StyleGAN2 (truncation = 1)
- StyleGAN2 (truncation = 0.7)
- Pix2Pix

Adjusting the StyleGAN2 *truncation* parameter balances visual quality against diversity. While higher truncation values yield more diverse results, they also increase the risk of generating image artifacts.

The graph shows that StyleGAN2 with truncation= 1.0 achieves almost identical performance to the Pix2Pix test set: the Kernel Density Estimation curves overlap.

The two GANs represent a clear trade-off between *geometric accuracy* (Pix2Pix) and *radiographic realism* (StyleGAN2).

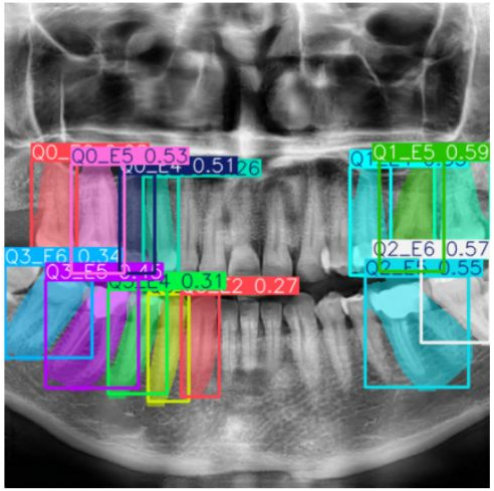
Confidence: Represents the statistical certainty of the estimate.
Density: Shows data concentration. The peak indicates the most frequent result.



Evaluation and Generalization study

	Avg Confidence	Wasserstein Dist.	FID Score
Real Images	83.64%	0.0000	0.0000
StyleGAN (trunc=1.0)	76.44%	0.0721	102.8314
StyleGAN (trunc=0.7)	77.94%	0.0570	119.9763
Pix2Pix	76.39%	0.0726	120.0621

Quantitative evaluation showing the trade-off between structural metrics (Avg Confidence/Wasserstein) and visual diversity (FID) across synthetic datasets.

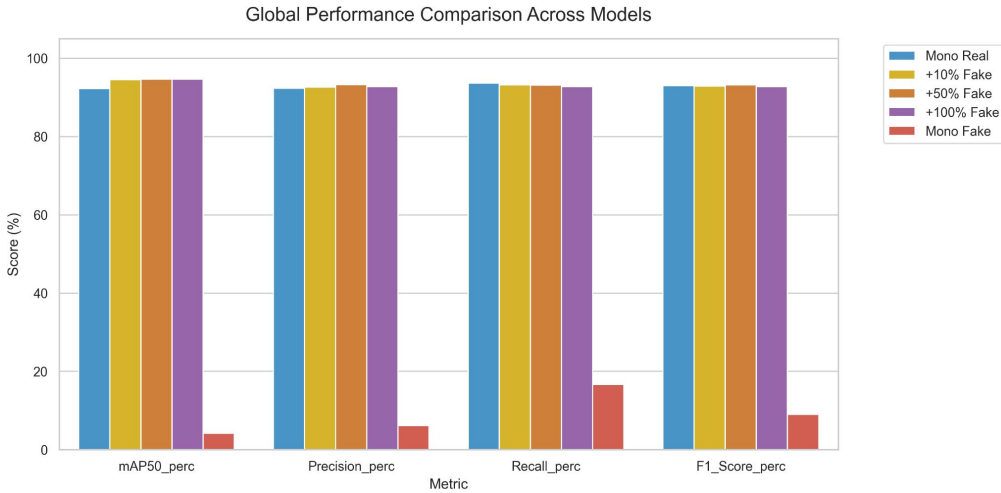


StyleGAN2 segmentation example
Teeth found: 14

- **Wasserstein Distance**: Measures the minimum "cost" required to transform one probability distribution into another.
- **Fréchet Inception Distance (FID)**: Evaluates the quality of generative models by comparing the statistical similarity between features of real and synthetic images.

Ablation study

	mAP50	Precision	Recall	F1-score
Mono Real	92.27	92.33	93.62	92.97
+10% Fake	94.53	92.62	93.20	92.91
+50% Fake	94.68	93.31	93.10	93.20
+100% Fake	94.63	92.73	92.80	92.76
Mono Fake	4.16	6.18	16.70	9.02



Quantitative evaluation under different data augmentation scenarios with 879 Pix2Pix synthetic images available.

Pix2Pix data augmentation **improves** YOLOv8 performance!

However, it will eventually undergo a performance drop once the real/fake image ratio exceeds optimality.

Possible improvements

Data & Pipeline Refinement

- **Pix2Pix Dataset:** Need for a better and more diverse dataset for training.
- **Mask Contours:** Better and more precise use of SAM for mask contour refinement. Implementing of MedSAM.

Hardware Limitations

- **Compute Power:** A better GPU with more VRAM would allow the training of better, SOTA models.
- **Time Constraints:** Need for longer training times, not restricted by platform limitations.

Strategic Use of Data Augmentation

- **Targeted Application:** Since general tooth segmentation is already highly effective, data augmentation would be more useful for the detection and segmentation of rarer, medically significant, and complex teeth structures that have smaller datasets.

References

IMAGE GENERATION

- Pano-GAN: A Deep Generative Model for Panoramic Dental Radiographs (2025)
- Evaluating the performance of generative adversarial networks for dental panoramic image synthesis (2023)
- PanoDiff-SR: Synthesizing Dental Panoramic Radiographs using Diffusion and Super-resolution (2025)
- Recent Advances in Dental Panoramic X-Ray Synthesis and Its Clinical Applications (2024)
- ToothMaker: Realistic Panoramic Dental Radiograph Generation via Disentangled Control (2023)
- High-fidelity medical image generation: controllable synthesis of high-resolution medical images via hierarchical fusion (2025)

SEGMENTATION

- YOLOv8: A Novel Object Detection Algorithm with Enhanced Performance and Robustness (2024)
- YOLO-DentSeg: A Lightweight Real-Time Model for Accurate Detection and Segmentation of Oral Diseases (2025)
- TransUNet: Transformers Make Strong Encoders for Medical Image Segmentation (2021)
- MICCAI STS 2024 Challenge: Semi-Supervised Instance-Level Segmentation (2024)